CAMPUS CANTEEN MANAGEMENT SYSTEM

USER MANUAL

Setup, customer workflows, administrator workflows, and project notes

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1. About the application

CCMS is a local terminal application for canteen accounts, menu browsing, ordering, inventory, customer feedback, and administrator reporting. This guide follows the current source files in the attached project. The administrator reporting module is present in source, but the configured build task currently omits it; see Section 3 before attempting a build.

2. Requirements and project layout

- Windows with the configured MSYS2 UCRT64 GCC toolchain is the project's documented environment.
- The application expects a console/terminal; color, screen clearing, and box characters depend on terminal support.
- Keep src/, lib/, and data/ together, including data/menu/ and data/reports/.
- Ensure data files and directories are writable for account, feedback, inventory, and report operations.

Project path	What it contains
src/	Application entry, account, menu/order, feedback, admin, and report modules.
lib/	Headers and support declarations, including the HTML-to-PDF interface.
data/menu/	Four category CSV files used by inventory browsing and ordering.
data/	Account, feedback, sales, and generated report data.
data/reports/	Generated report HTML and PDF output.

3. Build status and launch location

Build limitation

The current VS Code build task does not compile src/reports.c, even though src/admin_portal.c calls newreport() from that module. The task therefore fails during linking with an undefined reference to newreport. src/reports.c also calls html_to_pdf(), implemented in src/headless.c, which is not listed in the task. This documentation update does not change application/build code; the build task must be corrected and validated before it can build the report-enabled app.

The configured task uses MSYS2 UCRT64 GCC and writes cms.exe to the project root. Do not add the separate legacy src/home.c entry point to that build. The current task's exact source list is in .vscode/tasks.json.

Run from the data-relative working directory

Most active data accesses use paths such as ../data/menu/... and ../data/feedback.csv. Start the executable with src/ as the working directory so ../data resolves to the project's data/ folder. After a successful build, open a shell in the project src/ directory and run:

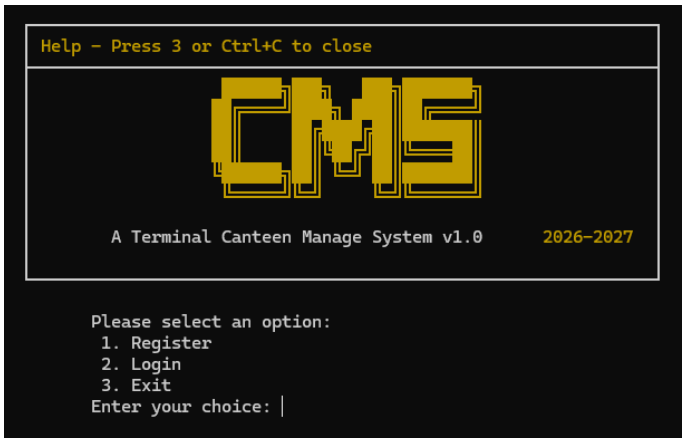
```
..\cms.exe
```

The VS Code task currently sets its build working directory to the project root; build location and runtime working directory are separate. Running from the project root can make ../data point outside the project.

ADMIN_FILE and CREDENTIALS_FILE environment variables override the admin and regular-user credential paths where those code paths support them.

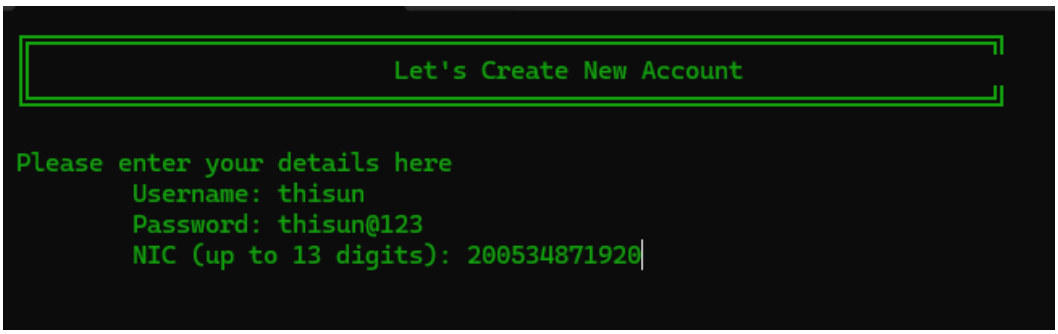
4. Accounts and sign-in

Main menu	Action
1. Register	Create a regular customer/student account.
2. Login	Sign in as a regular user or the reserved administrator.
3. Exit	Close the program.



Create a regular account

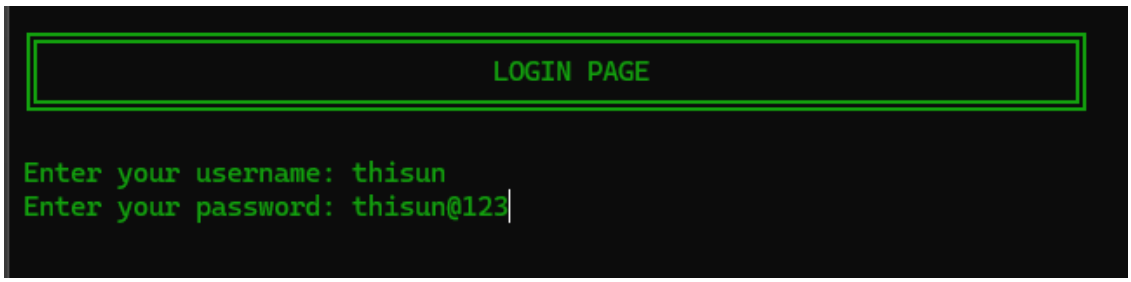
1. Choose Register from the main menu.
2. Enter a username containing letters, digits, or underscores. The admin username is reserved.
3. Enter a nonempty password. Commas and surrounding whitespace are rejected by the current validation.
4. Enter a NIC containing 1–13 digits.
5. The application saves the record to the credentials file. The current registration source does not check for duplicate usernames.



Sign in

Choose Login and enter the username and password. Regular usernames are compared without ASCII case sensitivity; passwords are case-sensitive. Password input is visible in the terminal.

The initial administrator username is admin and the source-defined first-use password is Admin@123. On first login, the app requests a new password. If your admin file already has a changed password, use that password instead.



5. Customer / student tasks

Browse inventory

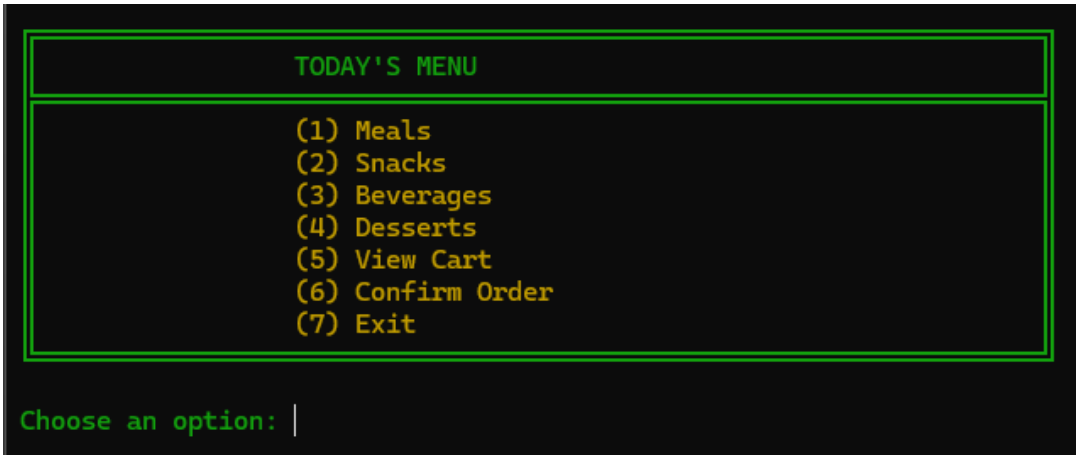
Choose Watch Inventory, select one of the four categories, and review item IDs, names, prices, and stock. Use the return prompt to leave the inventory view.



Place an order

6. Choose Buy Foods and select Meals, Snacks, Beverages, or Desserts.
7. Enter an item ID and a positive quantity. Add more items as desired.
8. Choose View Cart to check quantities and subtotal.
9. Choose Confirm Order. The application reloads menu data and checks availability before applying stock changes.
10. After the order is saved, review the receipt and confirmation message.

On confirmation, menu data is rewritten using temporary files and backups. The order flow appends item, quantity, unit price, and line total to ../data/reportdata.csv. If that append fails after stock is saved, the application reports the report error; inventory may already have changed.

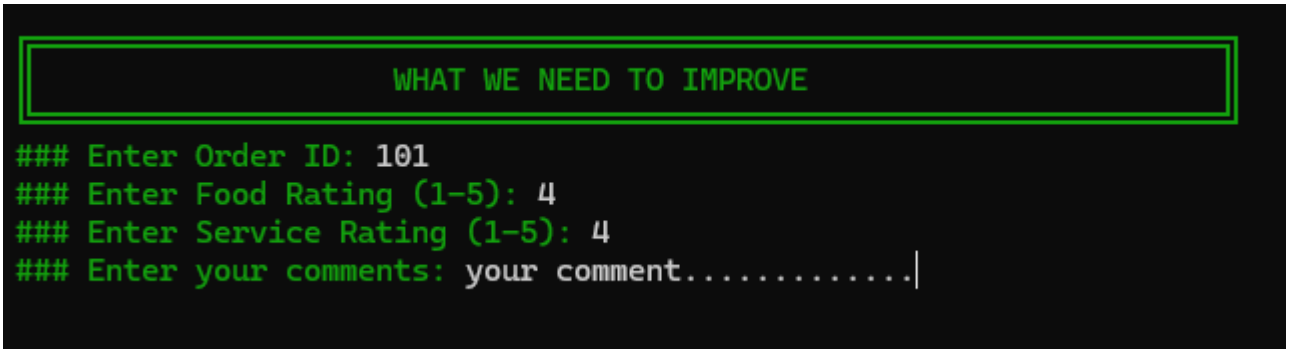


Submit feedback

11. Choose Give Feedback or Complain.
12. Enter a positive order ID and food and service ratings from 1 to 5.
13. Enter a comment and choose 1 to submit or 0 to cancel.

Order IDs are stored as entered and are not matched against completed orders.

The student menu's fourth option is marked under development.



6. Administrator tasks

Portal option	Purpose
1. View feedback	Display valid feedback records.
2. Calculate average ratings	Compute mean food and service ratings.
3. View complaints	Show records with a food or service rating of 2 or lower.
4. View inventory	Browse menu categories and stock.
5. View sales report	Read the generated data/reports.csv report.
6. Create new sales report	Generate reports.csv and report.html from data/sales.csv.
7. Exit to main menu	Return to the opening menu.

The option labels above reflect the current administrator portal source. The build issue in Section 3 means these report-enabled options are not available until the missing modules are linked.

Create/export a report

14. Use administrator option 6 after the report-enabled application has been built.
15. The report generator reads sales.csv and writes a tabular report to reports.csv and an HTML dashboard to reports/report.html.
16. The admin portal calls the report-generation function directly; it does not display the separate reports() routine's Y/N export prompt. The source contains a PDF export helper and target path, but that export is not connected to the admin portal flow.

The report input sales.csv is separate from reportdata.csv, which the order workflow appends to. The current source does not automatically merge those files. The report generator depends on sales.csv being available and populated.\

7. Files and reports

Path under data/	Description
admin.dat	Administrator record and first-login password-change flag.
credentials.csv	Regular account records.
menu/meals.csv	Meal items and stock.
menu/snacks.csv	Snack items and stock.
menu/beverages.csv	Beverage items and stock.
menu/desserts.csv	Dessert items and stock.
feedback.csv	Feedback records.
reportdata.csv	Order-flow sales lines.
sales.csv	Input for the separate report-generation workflow.
reports.csv	Generated administrator sales report.
reports/report.html	Generated visual report page.
reports/visual_report.pdf	Optional PDF export destination.

Do not move individual data files while the app is running. Back up the entire data/ directory before making manual changes.

8. Troubleshooting

Problem	What to check
Linker error: undefined reference to newreport	The VS Code task omits src/reports.c. The report module also uses html_to_pdf from src/headless.c.
Menu items cannot be opened	Confirm the current working directory is src/ and data/menu/ contains the category CSV files.
Registration cannot save	Check that ../data/credentials.csv is writable from the runtime working directory; confirm CREDENTIALS_FILE is not set to an invalid path.
Admin sign-in fails	Use the current admin password; check that the selected admin file contains a valid record and that ADMIN_FILE is not pointing elsewhere.
Feedback or reports appear empty	Confirm the app is using the expected data directory and that the relevant CSV contains valid records.
PDF export	The PDF helper exists, but the admin portal's report-generation option does not call the export prompt in the current flow.
Screen clearing or colors look wrong	Run in a compatible terminal and check its TERM/ANSI support.

9. Security and known limitations

This is a local learning/demo project, not production authentication or point-of-sale software. Regular and administrator passwords are stored in plain text, and password entry is visible. Do not use real or reused passwords or publish account files.

- No database, remote service, online payment, or multi-user locking is provided.
- Duplicate regular usernames are not checked in the visible registration flow.
- Feedback order IDs are not verified against purchases.
- Order reporting and administrator report generation currently use different input files.
- The configured VS Code task does not currently link the report module; successful full build has not been verified.